

Computer Vision Practical Report:

PERSPECTIVE TRANSFORMATION VIDEO

1. AIM

To read an input video stream or file using OpenCV, apply Perspective Transformation frame-by-frame using a calculated transformation matrix, display the original and transformed video frames simultaneously, and save the modified video to disk.

2. PROCEDURE

- Open the input video stream using `cv2.VideoCapture()`.
- Retrieve video properties such as frame width, height, and FPS.
- Define four source points on the video frame and four corresponding destination points for the output view.
- Calculate the 3×3 perspective transformation matrix using `cv2.getPerspectiveTransform()`.
- Initialize a video writer using `cv2.VideoWriter()` to save the output video.
- Read each frame sequentially in a loop and apply the perspective transformation using `cv2.warpPerspective()`.
- Display the original and transformed frames in real-time, write the processed frames to disk, and release all video resources upon completion.

3. PROGRAM CODE

```
import cv2
import numpy as np

# Read video
cap = cv2.VideoCapture("E13-VID.mp4")

# Check if video is opened
if not cap.isOpened():
```

```
print("Error: Cannot open video")
exit()

# Get video properties
width = 500
height = 400
fps = cap.get(cv2.CAP_PROP_FPS)

# Define output video
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter(
    "perspective_output.mp4",
    fourcc,
    fps,
    (width, height)
)

# Define four points in the original frame
pts1 = np.float32([
    [50, 50],
    [450, 50],
    [50, 350],
    [450, 350]
])

# Define corresponding points
pts2 = np.float32([
    [80, 80],
    [420, 50],
    [50, 350],
    [450, 300]
])

# Calculate Perspective Transformation Matrix
M = cv2.getPerspectiveTransform(pts1, pts2)
```

```
while True:
    ret, frame = cap.read()

    if not ret:
        break

    # Resize frame to small size
    frame = cv2.resize(frame, (width, height))

    # Apply perspective transformation
    transformed = cv2.warpPerspective(
        frame, M, (width, height)
    )

    # Display output
    cv2.imshow("Perspective Transformed Video", transformed)

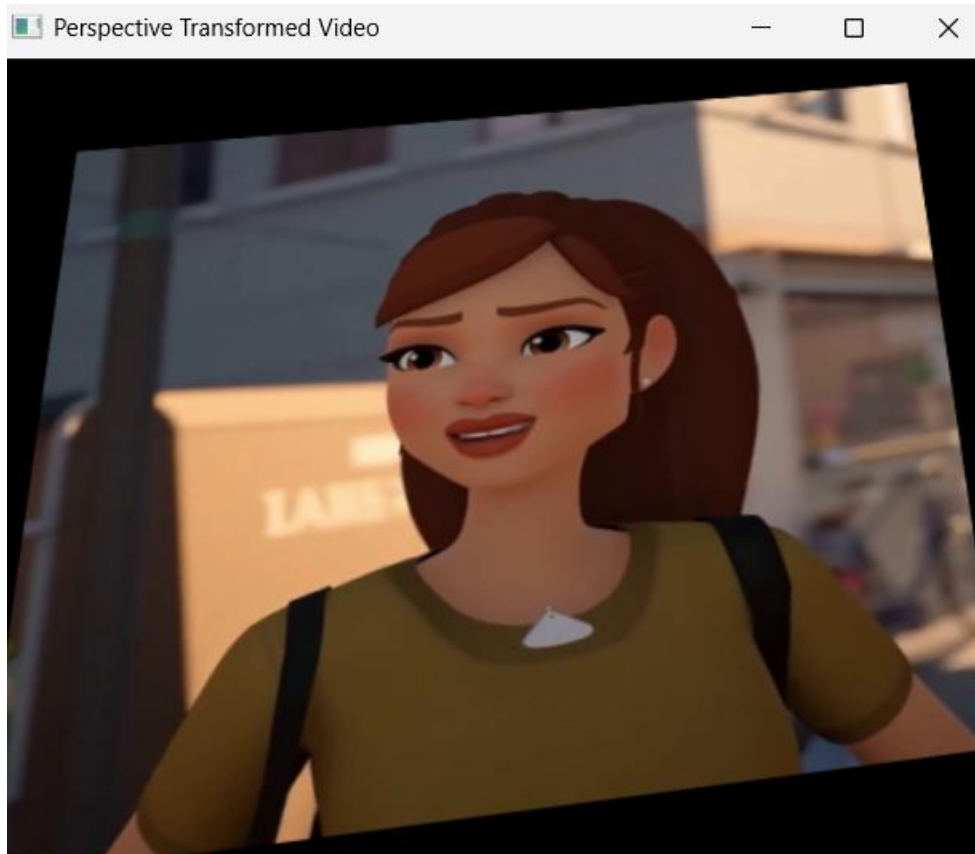
    # Save transformed frame
    out.write(transformed)

    # Press 'q' to stop
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

# Release resources
cap.release()
out.release()
cv2.destroyAllWindows()

print("Perspective transformation completed!")
print("Output saved as perspective_output.mp4")
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.